1. Title : Modified Silica Filler for Space Application

Brief Description and Product/Solution

Modified Silica Filler for Space Application is a high temperature crystalline polymorph of silica. Modified silica filler is used as reinforcement filler in the adhesive used for bonding E-glass, strain isolation pads and ceramic tiles to space vehicles. C-MET has developed a technology for the production of space qualified phase pure modified silica filler in pilot plant scale. ISRO used modified silica filler supplied by C-MET in their Space capsule recovery experiments. C-MET has so far supplied 440kg of modified silica to ISRO.



Modified Silica Filler for Space Application

Main uses and domain

Reinforcing filler for adhesives for space applications. It can also find applications in Structural paints, Decorative plaster works, Road marking paints, Engineered stone, Antiblocking additive in polymer films like PP & PE films, High wear coatings (varnishes, clear coats, gel coats), Marine coatings/paints, Faltting agent (clear coats, architectural and powder coats), Pigment extender (paints, caulks, powder coatings), Filler (elastomers, PET, sealants, caulks, resins), Abrasives (polishing), Quartz countertops, Reinforcement for quartz glass, Investment casting (Dental, jewellery, high precision metal casting), Sealant in dental moulding, Anti corrosion paints, Infrared absorber for transparent polymer films (for vegetable cultivation and ware house), Artificial marble, Filler for high quality polymer concrete, as filler in plastic based scratch and boiling water resistant sanitary wares, etc.

Features and Technical Specifications

<u>Specifications of the Modified silica filler</u>

1. Material	: Modified crystalline silica
2. Phase purity	: Phase pure (Tested by XRD)
3. Average particle size	: < 8 microns
4. Density	: 2.3 g/cc

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